

Posterior-Predictive Pass-at- k

a two-parameter replacement for the independence extrapolation in fine-tuning diagnostics

Peter Cotton

September 2, 2026

Abstract

Pass-at- k diagnostics for post-training frequently extrapolate from a per-prompt success estimate $\hat{p}$ by the independence formula $1 - (1 - \hat{p})^k$; the coverage-ratio diagnostic of [Malladi et al. \(2026\)](#), for instance, converts pass@1 to pass@16 by $f_{16}(p) = 1 - (1 - p)^{16}$. We measure what the plug-in costs on released per-sample rollout records (nine thousand prompts, eight scored samples each) and replace it with the posterior-predictive quantity $E[1 - (1 - p)^k \mid s]$ under a two-parameter empirical-Bayes difficulty prior fitted by marginal maximum likelihood. Scoring held-out samples, the plug-in attains log loss 3.37 per prompt, driven by prompts with no training successes that it declares impossible and that then succeed; the posterior predictive attains 0.48. Extrapolating the aggregate curve from half the samples, the plug-in understates pass@8 by thirteen points absolute while the posterior predictive lands within two. Probit-normal and beta-binomial priors tie to three decimals, so nothing hangs on the family. The bias shrinks as the per-prompt sample count grows, and the study covers one model on one environment suite; both limits are stated where they bind.

1 Introduction

A language model is sampled k times on a prompt and succeeds if any sample passes; pass@ k , the probability of that event averaged over prompts, is the working currency of post-training evaluation. When $n \geq k$ scored samples per prompt are available, the unbiased estimator of [Chen et al. \(2021\)](#), $1 - \binom{n-c}{k} / \binom{n}{k}$ with c the correct count, settles the matter. The trouble begins when k exceeds the samples at hand and the curve must be extrapolated. The common device is the independence formula: estimate a per-prompt success rate $\hat{p}$, report $1 - (1 - \hat{p})^k$. It appears inside diagnostics that steer training decisions; [Malladi et al. \(2026\)](#) write “we convert pass@1 to an estimated pass@16 value using $f_{16}(p) := 1 - (1 - p)^{16}$ ” as a step of a coverage-ratio rule that decides where their filtered fine-tuning method should be applied, and the rule works well in their experiments.

This note is about the extrapolation step, not the diagnostic built on it. The formula is exact when p is known; applied to a noisy $\hat{p}$ it inherits a specific pathology. A prompt with zero successes among m training samples gets $\hat{p} = 0$ and a predicted pass@ k of zero at every k , a confident impossibility claim that a latent success rate of, say, 0.1 will refute given enough samples. We measure the cost of the plug-in on released rollout records and show that the two-parameter empirical-Bayes repair, a posterior expectation in place of a plug-in evaluation, removes most of it.

2 The plug-in and its failure mode

Let a prompt carry a latent per-sample success probability p , and let $s \sim \text{Bin}(m, p)$ successes be observed among m training samples. The plug-in predictor of $\text{pass}@k$ is

$$\widehat{\text{pass}}_k^{\text{plug}} = 1 - \left(1 - \frac{s}{m}\right)^k. \quad (1)$$

Two defects follow from the noise in s/m . At $s = 0$ the prediction is exactly zero however small m is, so every eventual success on such a prompt costs unbounded log loss. And since $(1 - p)^k$ is convex in p , Jensen’s inequality gives $E(1 - \hat{p})^k \geq (1 - E\hat{p})^k$, so the plug-in curve is biased downward in aggregate once $\hat{p}$ is noisy, increasingly so as k grows.

3 The posterior predictive

Give p a prior π_θ with two parameters fitted by marginal maximum likelihood across prompts, and predict with the posterior expectation

$$\widehat{\text{pass}}_k^{\text{post}} = E[1 - (1 - p)^k \mid s]. \quad (2)$$

Two convenient families serve. Under a $\text{Beta}(a, b)$ prior the posterior is $\text{Beta}(a + s, b + m - s)$ and the moment is a falling product,

$$E[(1 - p)^k \mid s] = \prod_{j=0}^{k-1} \frac{b + m - s + j}{a + b + m + j}, \quad (3)$$

so Equation (2) is closed-form. Under a probit-normal prior, $p = \Phi(\theta)$ with $\theta \sim N(\mu, \tau^2)$, the marginal likelihood of the observed counts and the posterior moment are one-dimensional Gauss–Hermite quadratures over θ ; this is the family that composes with correlated extensions, since a factor-correlated portfolio of successes conditions to exactly this form (Cotton, 2026). In the exchangeable setting of this note the two families are competitors on equal footing, and the study below treats them as such.

4 Numerical study on released rollouts

The data are the publicly released Pass8 rollout records of the RLVE training suite: nine thousand prompts across eighteen procedurally generated environments, eight scored samples per prompt from Qwen3-4B-Thinking, three gigabytes of response text reduced to a forty-one-kilobyte outcome table.¹ Samples 0–3 of each prompt are the training half; samples 4–7 are held out. The held-out per-prompt $\text{pass}@4$ is then a single Bernoulli outcome, whether any of the four succeeded, so predicted probabilities are scored by log loss and Brier score with no estimator ambiguity. The fitted priors are $\text{Beta}(0.40, 0.83)$ and probit-normal $\mu = -0.78$, $\tau = 1.39$; both put substantial mass near $p = 0$ and $p = 1$, which is what the environments look like.

The per-prompt comparison in Table 1 is seven-fold in log loss and twenty percent in Brier, for two fitted parameters. The aggregate comparison in Table 2 shows the plug-in bias growing with the extrapolation distance exactly as the convexity argument says it must.

¹Dataset: CL-From-Nothing/RLVE-Qwen3-4B-Thinking-2507-Pass8-Rollouts on Hugging Face. Extraction and experiment scripts are committed under `research/cavity_calculus/exp2_passk/` in the `winning` repository.

predictor of held-out pass@4	log loss	Brier
plug-in $1 - (1 - s/4)^4$	3.367	0.192
posterior predictive, probit-normal	0.483	0.160
posterior predictive, beta-binomial	0.483	0.160

Table 1: Per-prompt scores over nine thousand held-out Bernoulli outcomes. The plug-in’s log loss is dominated by zero-training- success prompts that succeed in the held-out half; the posterior predictive prices those prompts at their shrunken posterior rate instead of zero. The two prior families tie.

aggregate	$k = 1$	$k = 2$	$k = 4$	$k = 8$
truth (all eight samples)	0.322	0.442	0.569	0.682
plug-in from the training half	0.322	0.412	0.495	0.549
probit-normal posterior	0.323	0.444	0.561	0.663
beta-binomial posterior	0.323	0.444	0.560	0.661

Table 2: Aggregate pass@ k extrapolated from four samples per prompt, against the unbiased estimator of [Chen et al. \(2021\)](#) on all eight. The plug-in’s downward Jensen bias grows with k and reaches thirteen points absolute at $k = 8$; the posterior predictives land within two points.

5 Scope

The plug-in bias scales with the noise in $\hat{p}$, hence shrinks as the per-prompt sample count grows; the study uses four training samples per prompt, and [Malladi et al. \(2026\)](#) do not state the count behind their pass@1 estimates, so no claim is made about the magnitude in their setting, only about the mechanism and its two-parameter repair. The data cover one model on one environment suite; transfer to math and code benchmarks is plausible and unshown. The two prior families are statistically indistinguishable here, so the choice between them can be made on convenience or, when correlated portfolios of attempts arise, on composability ([Cotton, 2026](#)). Diagnostics built on the plug-in, including the coverage ratio of [Malladi et al. \(2026\)](#), accept the posterior predictive as a drop-in for their extrapolation step, and measuring that substitution in their setting is the natural next experiment.

References

- M. Chen et al. Evaluating large language models trained on code. arXiv:2107.03374, 2021.
- P. Cotton. Scalable inversion of contests with correlated performances, including softmax and multinomial probit. arXiv:2609.01133, 2026.
- S. Malladi, S. Jelassi, D. Foster, J. T. Ash, and A. Krishnamurthy. TailSFT: filtered fine-tuning improves post-training performance. arXiv:2608.25756, 2026.